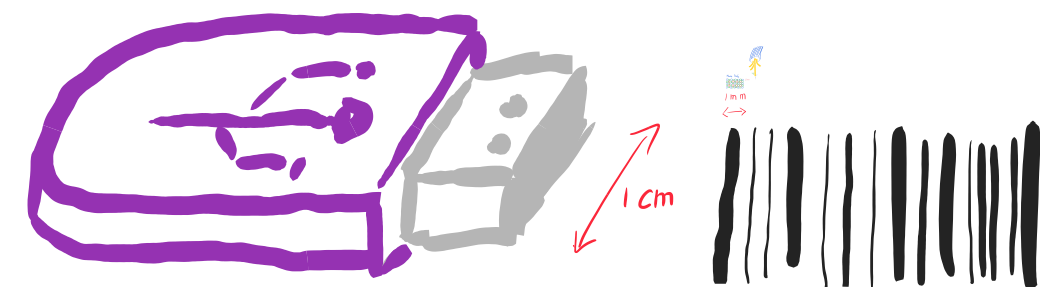


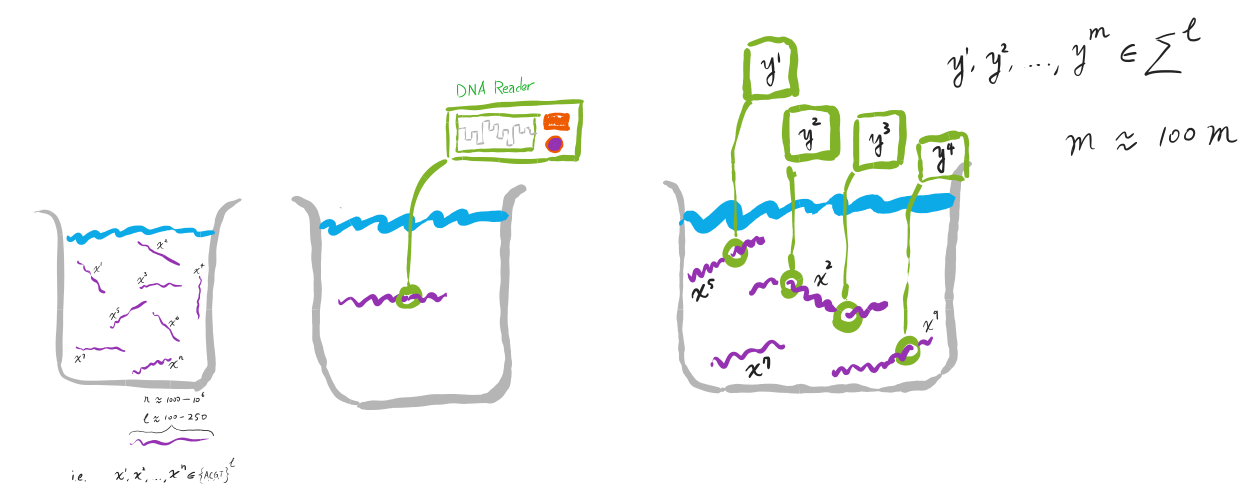
Why DNA For Data Storage?



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GENO-WEAVING:
INTERLEAVING POLAR CODES
ACROSS STRANDS
FOR DNA DATA STORAGE

Implementation Details



Mathematical Problem Statement

n strands $n \approx 1000-10^4$
 ℓ letters per strand $\ell \approx 10^2-250$
 $x^1, x^2, x^3, \dots, x^n \in \{A, C, G, T\}^\ell$

Read m times, $m \approx 100n$
 $y^i = x^{H(i)} + \text{noise}$
 $H: [m] \rightarrow [n]$ random function
 "Sample with Replacement"

JOB:
 Recover $\{x^1, x^2, \dots, x^n\}$
 (unknown set)
 using $y^1, y^2, \dots, y^m$

Capacity Results
 $I(x^1, \dots, x^n; y^1, \dots, y^m, H)$
 $\parallel$
 $n \cdot \ell \cdot \text{Capacity}(W^{\otimes \lambda})$
 $\lambda = \frac{m}{n}$

$I(x^1, \dots, x^n; y^1, \dots, y^m)$
 $\parallel$
 $n \cdot \ell \cdot \text{Capacity}(W^{\otimes \lambda}) - \log_2 \left(\frac{m}{n} n! \right)$

This Talk $\rightarrow$

Future? $\rightarrow$

USE Polar Codes

Can't we just use any capacity-achieving codes?

Polar Codes can work with ANY input distribution

- How to build the decoder?
- How to build the encoder?
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